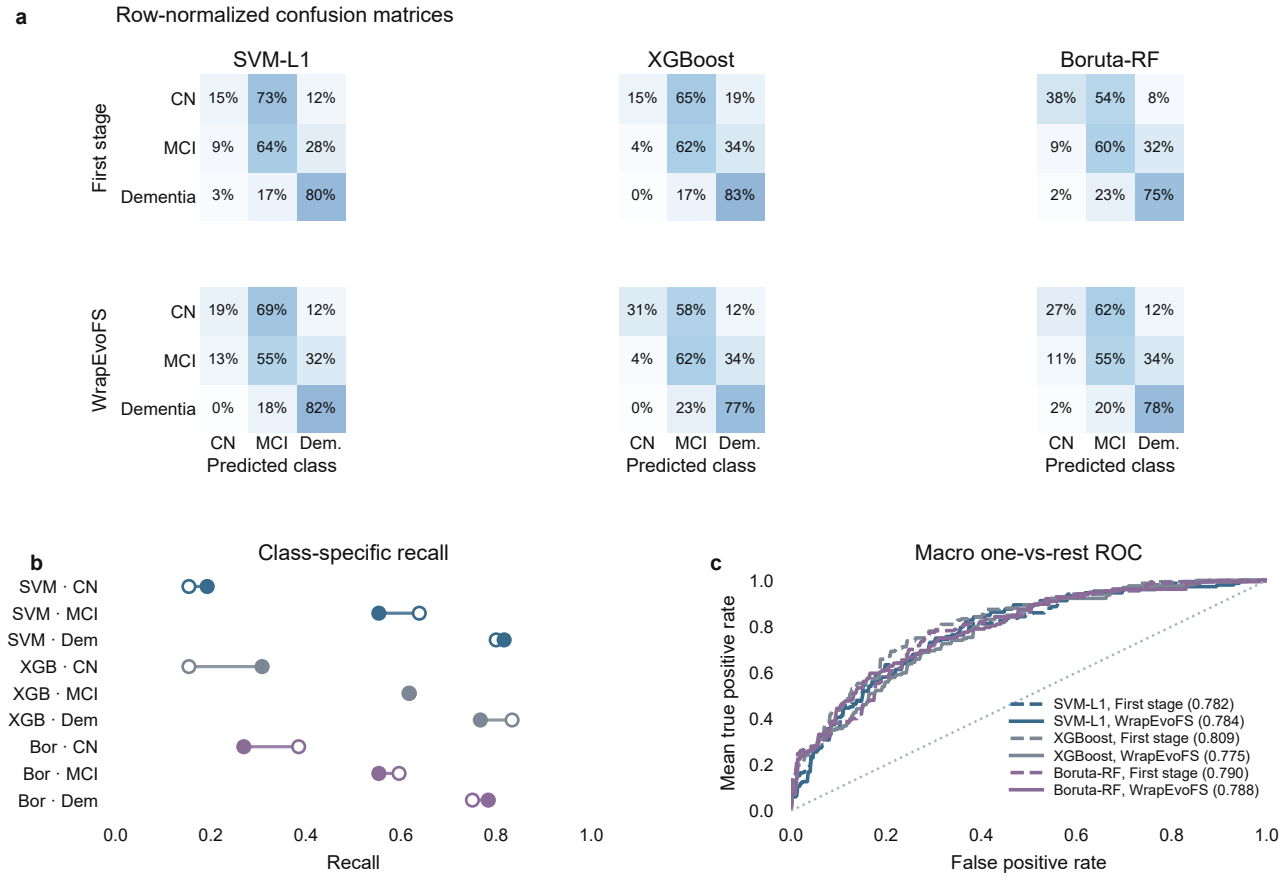

a
Row-normalized confusion matrices
SVM-L1
XGBoost
Boruta-RF
First stage
WrapEvoFS
CN
MCI
Dementia
15% 73% 12%
9% 64% 28%
3% 17% 80%
15% 65% 19%
4% 62% 34%
0% 17% 83%
38% 54% 8%
9% 60% 32%
2% 23% 75%
19% 69% 12%
13% 55% 32%
0% 18% 82%
31% 58% 12%
4% 62% 34%
0% 23% 77%
27% 62% 12%
11% 55% 34%
2% 20% 78%
CN MCI Dem.
Predicted class
b
Class-specific recall
SVM · CN
SVM · MCI
SVM · Dem
XGB · CN
XGB · MCI
XGB · Dem
Bor · CN
Bor · MCI
Bor · Dem
0.0 0.2 0.4 0.6 0.8 1.0
Recall
c
Macro one-vs-rest ROC
Mean true positive rate
False positive rate
SVM-L1, First stage (0.782)
SVM-L1, WrapEvoFS (0.784)
XGBoost, First stage (0.809)
XGBoost, WrapEvoFS (0.775)
Boruta-RF, First stage (0.790)
Boruta-RF, WrapEvoFS (0.788)